

Machine learning for code smell detection: a systematic review

Abstract

Code smells indicate potential design problems in software. The goal of this systematic review is to characterize machine learning approaches for automated code smell detection and to identify open research challenges.

3. Review protocol

Following Kitchenham's guidelines, we defined the search string ("code smell" OR "bad smell" OR anti-pattern) AND ("machine learning" OR classification OR "deep learning") and executed it on IEEE Xplore, the ACM Digital Library and Scopus. We additionally performed backward snowballing on the reference lists of all included papers. Peer-reviewed studies published between 2005 and 2023 were eligible; grey literature was excluded.

4. Results

The search retrieved 1,286 candidate papers, of which 74 primary studies were retained after full-text screening.